

Enhancing UAV Control and Features with AI Integration



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INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are playing an increasingly vital role in the low-altitude economy, supporting applications such as search-and-rescue, last-mile delivery, and real-time monitoring. Hexacopters, in particular, offer improved payload capacity and flight stability, making them ideal for mission-critical deployments. As demand grows for more intelligent and autonomous UAV systems, integrating advanced flight control and onboard AI becomes essential.

OBJECTIVE

This project rebuilds a disassembled UAV by equipping it with a Pixhawk 6X flight controller and integrating AI-powered object detection using Jetson Nano. The goal is to develop a smart UAV capable of supporting low-altitude economy missions. Additionally, the system is tested in experimental scenarios for marine life observation and participates in a heterogeneous multi-robot system, collaborating with ground or underwater robots in coordinated tasks.

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Hexacopter in Hover Operation



Major Hardware Components

- Pixhawk 6X Flight Controller**
Provides high-performance flight control with advanced IMU and sensor integration
- NVIDIA Jetson Orin Nano Super**
Runs real-time AI models for object detection and tracking onboard the UAV.
- M9N GPS Module & Telemetry System**
Enables autonomous navigation and remote monitoring.
- Carbon Fiber Drone Frame**
A durable and lightweight frame adapted from a previous prototype.
- AIR2216 KV920 CW Brushless Motors & Electronic Speed Controllers**
Ensures responsive and stable propulsion.
- 4S 5300mAh Li-Po Battery Pack**
Supplies power to both propulsion and onboard computing systems.
- 4K Wide Angle Camera Module**
Captures live video for computer vision tasks such as target detection and tracking.

METHODOLOGY - CONTROL OPTIMIZATION

The UAV is equipped with a Pixhawk 6X flight controller running ArduPilot firmware. Setup involves calibrating the IMU, GPS, ESCs, and radio via Mission Planner. PID tuning—both manual and autotune—is performed to optimize flight stability and hover accuracy.

The controller adjusts roll, pitch, and yaw using the PID formula:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where $u(t)$ is the control signal, and $e(t)$ is the error between desired and actual angles.

Test flights in controlled conditions allow iterative tuning, with performance refined through log-based feedback.

RESULT

Metric	Simulated (eCalc)	Tested (Real)
Hover Time	~13.6 min	~20 min
Power at Hover	~264 W	~230–240 W (est.)
Hover Throttle	32%	~30–35%
T/W Ratio	2.9:1	2.9:1

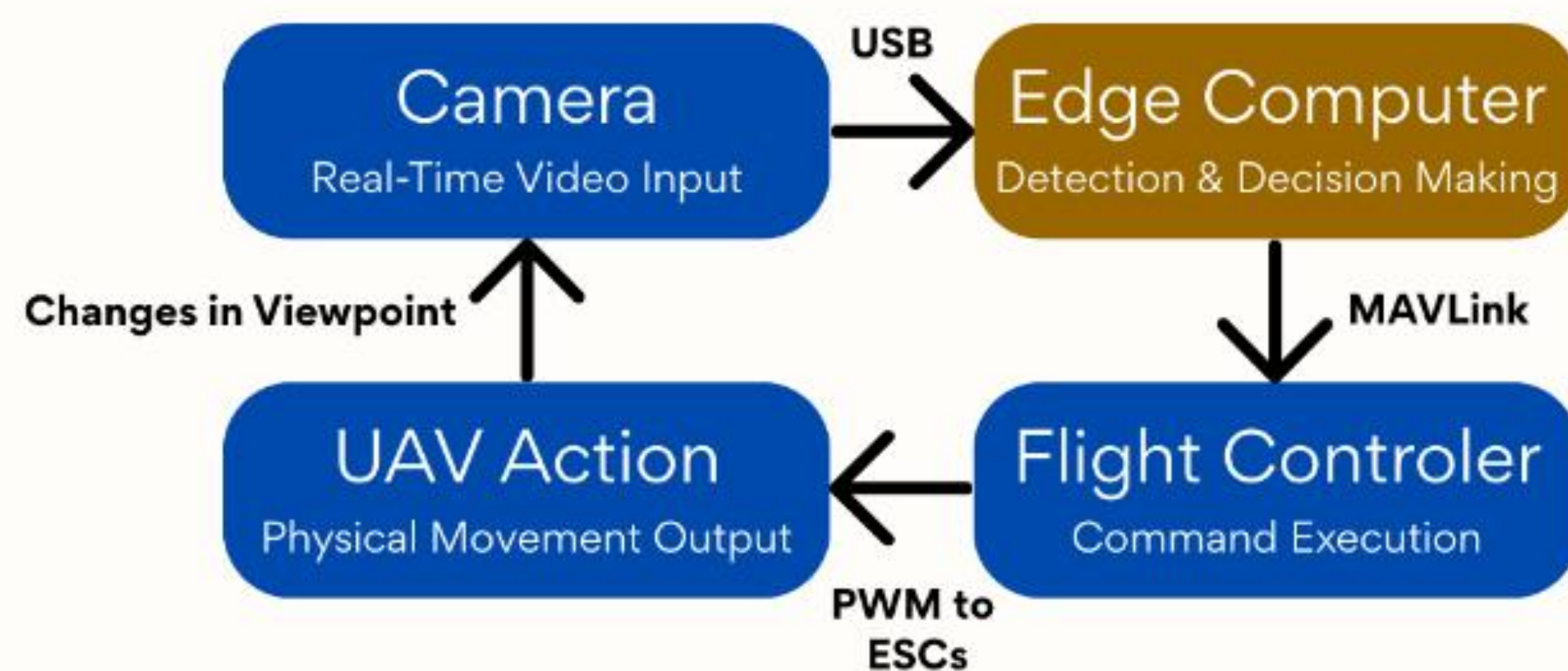
With optimized PID tuning, the UAV achieved up to 20 minutes of hover time at 2.2 kg, significantly outperforming the ~13.6-minute simulated estimate. Power consumption was reduced to 230–240W, allowing a projected flight range of 3.4km—comparable to a round trip between HKUST and Tseung Kwan O Pier. These results demonstrate the UAV’s readiness for practical deployment.

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METHODOLOGY- AI INTEGRATION

The UAV features onboard edge computing powered by the Jetson Orin Nano Super, enabling real-time AI inference during flight. A camera module captures live video, which is processed on-device for object detection and tracking using a YOLOv8 model. To ensure optimal performance, the YOLOv8 model is converted into a TensorRT .engine file, allowing fast, low-latency inference on the Jetson GPU.

The AI control system follows the pipeline below:



This closed-loop system allows the UAV to detect, decide, and act autonomously, with no dependency on ground stations or external computation.

RESULT

The YOLOv8 model runs in real time at ~40 FPS on 720p video using the Jetson Orin Nano in 15W mode. It reliably detects common objects, especially persons, even when only partially visible in the frame—making it well-suited for human-following applications. However, in windy conditions, image instability can lead to detection errors or loss of tracking. This may cause false movement commands, requiring occasional manual intervention.

Edge Computer Setup



Real time object detection



EXPERIMENT

Extensive test flights were first conducted at the grass field outside the USEL Lab near Shaw Auditorium to verify basic flight control, object detection stability, and system safety in a controlled environment.

Following successful preliminary tests, a formal flight experiments were carried out at the HKUST Pier, a coastal outdoor site with naturally windy conditions. The final tests were conducted with official permission from the HKUST Security Center and the Water Sports Center, following all necessary safety precautions.

The mission objectives included validating flight stability under outdoor wind disturbances and evaluating the real-time detection and tracking of a stationary human target and a navigating Unmanned surface vehicle (USV). The AI consistently recognized the USV as a “boat” and the human target as a “person”. The UAV demonstrated reliable object detection, with occasional tracking loss in strong wind conditions.

CONCLUSION

Mission Readiness for Low-Altitude Economy

The developed UAV platform demonstrates strong potential for low-altitude economy applications, offering stable hover performance (up to 20 minutes at 2.2kg), precise flight control, and real-time AI detection at ~40 FPS. These features make it well-suited for tasks such as autonomous delivery, search-and-rescue, and environmental monitoring, even in dynamic or GPS-limited environments.

Research Flexibility and Algorithm Integration

Its modular architecture supports rapid deployment of custom Python-based AI models and enables advanced tasks such as object tracking, person-following, and UAV-USV collaboration. The system is compatible with offboard MAVLink control and can be extended with ROS 2 for more complex, multi-robot missions involving distributed autonomy.

Scalability for Future Expansion

With support for sensors like LiDAR, RTK, Intel RealSense, and Mid360, the platform is highly extensible. Powered by Jetson Orin Nano and open-source firmware, it provides a robust foundation for future research in SLAM, swarm coordination, and onboard autonomy.

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If you're interested in this project or potential collaboration, feel free to reach out.
We are actively looking for research partners and development opportunities.

USV Monitoring Mission

